

Agentic AI Testing Platform — PRD & Design Document

Codename: Spec-1 A11y (Phase-1 of AI Test Engineer Platform) Environment: 100% Replit (Deployments + Postgres + Upstash Redis) Goal: Build a multi-tenant agentic testing platform for Accessibility Testing (WCAG 2.2). The platform will crawl, audit, deduplicate, and report accessibility issues with Vially-style UI.

Core Features

- Multi-tenant Auth (Google/GitHub/Email) • Page Crawler (robots.txt aware) • A11y Agent (Lighthouse + axe-core) • Evidence capture and Rollups • Reports (CSV/PDF) • Integrations: Jira, Slack • RBAC & Org isolation

System Architecture

Monorepo (apps/web-ui, services/api, workers/*, packages/*) Infra: Postgres (Neon/Supabase), Redis (Upstash), Cloudflare R2, Replit Deployments Flow: User → UI → API → Queue → Workers → DB → Report → UI

Database Schema (Key Tables)

User, Organization, Membership, Project, Estate, Page, A11yResult, A11yRollup, ApiKey, AuditLog. All queries scoped by orgId. Managed by Prisma ORM with Postgres.

Authentication & Security

Auth.js (NextAuth) with JWTs, RSA keys via JOSE. RBAC roles: OWNER, ADMIN, DEV, VIEWER. API keys (Argon2id), Fastify rate limit (Redis). Audit logs for compliance.

Agent Design

CrawlAgent → discovers pages. AuditAgent → runs Lighthouse + axe-core. DeDupeAgent → merges templates. ReportAgent → aggregates and exports. TicketAgent → pushes Jira issues.

UI Design

Next.js + Tailwind + shadcn/ui + Recharts. Pages: Dashboard, Issues Table, Reports, Integrations, Admin. Responsive, accessible, Vially-like interface.

Performance & Scaling

Replit Deployments for UI/API/workers. Neon autoscaling Postgres, Upstash Redis, R2 for storage. Crawl budget ≤ 100 pages, retries $\times 3$, job resume checkpoint.

Milestones

M0 – Foundation: Auth + DB (2 wks) M1 – Crawl/Audit: Playwright + axe/Lighthouse (2 wks) M2 – UI/Reports: Dashboard + Export (2 wks) M3 – Integrations: Jira/Slack (1 wk) M4 – Hardening: Security & Rate limit (1 wk)

Tech Stack Summary

Frontend: Next.js 14 + Tailwind + shadcn/ui Backend: Fastify + Prisma + Auth.js + BullMQ Workers: Playwright + Lighthouse + axe-core DB: Postgres (Neon) Queue: Redis (Upstash) Storage: R2/S3 Deploy: Replit

Future Enhancements

AI Test Engineer (v2), Functional & API Testing, Security Agent, IDE Plugin, Billing.

Acceptance Criteria

■ Login, create org/project/estate ■ Crawl pages, audit results stored ■ Dashboard & CSV/PDF exports ■ Jira integration ■ RBAC + Rate limiting